

MADRID-BARAJAS AP, Spain

WMO: 082210

Lat: 40.467N

Lon: 3.556W

Elev: 582

StdP: 94.53

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.5	-2.1	-10.0	1.7	3.9	-7.5	2.1	4.5	11.7	9.9	10.3	9.1	1.6	350	0.473

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	36.8	18.4	35.5	18.1	34.1	17.6	20.6	32.5	19.5	31.4	18.7	30.7	4.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.5	24.3	15.5	11.8	23.1	14.6	11.1	22.4	61.6	32.5	57.9	31.3	55.1	30.8	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.4	7.4	DB	-6.2	39.1	1.8	1.2	-7.5	40.0	-8.5	40.7	-9.5	41.4	-10.8	42.3	(4)
(5)			WB	-6.9	21.4	1.9	1.7	-8.2	22.7	-9.3	23.7	-10.4	24.6	-11.7	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.0	6.1	7.3	10.4	12.9	17.0	22.5	25.6	25.3	20.6	15.6	9.7	6.4	(6)	
	DBStd	7.57	2.80	2.71	2.87	3.19	3.63	3.54	2.78	2.49	3.06	2.97	3.13	2.93	(7)	
	HDD10.0	407	124	82	30	9	1	0	0	0	0	2	43	116	(8)	
	HDD18.3	1909	379	309	245	165	68	7	1	0	13	92	261	369	(9)	
	CDD10.0	2233	3	6	43	96	218	376	485	473	319	177	33	5	(10)	
	CDD18.3	692	0	0	0	1	27	132	227	214	82	8	0	0	(11)	
	CDH23.3	9351	0	0	4	66	430	1814	3136	2796	972	133	0	0	(12)	
	CDH26.7	4649	0	0	0	5	133	889	1748	1489	362	23	0	0	(13)	
(14) Wind	WSAvg	2.9	2.7	3.0	3.2	3.3	3.0	3.0	3.0	2.9	2.7	2.5	2.7	2.4	(14)	
(15) Precipitation	PrecAvg	384	31	31	30	40	44	20	7	8	24	61	49	38	(15)	
	PrecMax	546	95	94	94	95	105	60	34	29	66	158	182	126	(16)	
	PrecMin	239	0	1	0	11	0	1	0	0	0	8	3	2	(17)	
	PrecStd	80	25	21	22	21	30	17	8	9	16	37	38	36	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.2	19.5	23.8	27.3	32.7	38.0	38.7	37.9	35.0	29.4	21.7	16.1	(19)	
		MCWB	10.1	10.6	12.4	14.8	16.5	18.5	19.0	18.3	18.1	15.7	13.4	10.9	(20)	
	2%	DB	14.2	17.1	21.9	25.3	30.1	35.3	37.0	36.3	32.2	26.8	19.1	14.2	(21)	
		MCWB	9.4	9.8	11.7	13.9	15.8	18.0	18.6	18.2	17.0	15.1	12.7	10.3	(22)	
	5%	DB	12.7	15.1	20.0	23.2	27.8	33.5	35.8	35.1	30.4	24.6	17.1	12.9	(23)	
		MCWB	8.8	9.2	11.1	13.3	15.1	17.3	18.3	18.0	16.5	14.6	11.7	9.7	(24)	
	10%	DB	11.2	13.3	18.0	20.8	25.7	31.8	34.3	33.6	28.8	22.5	15.2	11.6	(25)	
		MCWB	8.0	8.5	10.3	12.0	14.5	16.9	17.7	17.6	16.1	14.1	11.0	8.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	12.5	14.0	15.9	18.2	21.0	22.5	21.8	19.7	17.8	15.1	12.9	(27)	
		MCDB	14.5	16.3	20.9	24.8	28.5	33.0	35.5	33.8	29.7	24.8	19.0	14.3	(28)	
	2%	WB	10.6	11.1	12.7	14.7	16.9	19.4	20.8	20.5	18.7	16.7	13.6	11.4	(29)	
		MCDB	12.9	15.2	19.7	23.7	26.9	31.4	33.3	32.2	28.7	22.7	17.3	13.3	(30)	
	5%	WB	9.5	9.9	11.8	13.6	16.0	18.5	19.4	19.4	17.8	15.9	12.6	10.2	(31)	
		MCDB	11.8	14.0	18.1	21.7	25.3	30.7	32.4	31.0	27.2	21.3	15.9	12.2	(32)	
	10%	WB	8.5	8.9	10.8	12.6	15.1	17.7	18.5	18.5	17.0	15.1	11.6	9.2	(33)	
		MCDB	10.8	12.6	16.5	19.3	23.6	29.3	31.5	30.5	25.8	20.3	14.4	11.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	11.1	12.4	12.1	13.1	14.9	15.6	15.3	13.8	11.8	9.8	9.4	(35)	
		MCDBR	12.6	15.3	17.6	17.0	17.5	17.7	17.1	17.2	16.8	16.0	13.0	11.4	(36)	
	5% WB	MCWBR	8.2	8.9	9.2	7.9	6.7	5.9	6.1	6.4	6.5	7.2	7.4	7.7	(37)	
		MCWBR	9.3	12.2	14.7	15.0	14.8	15.7	16.1	15.5	14.0	11.7	10.2	8.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.292	0.306	0.341	0.350	0.364	0.367	0.349	0.359	0.348	0.334	0.306	0.291	(40)	
		taud	2.508	2.478	2.398	2.381	2.368	2.380	2.403	2.381	2.423	2.477	2.512	2.517	(41)	
	Ebn at Noon		879	915	912	923	913	906	920	902	892	868	852	850	(42)	
		Edh at Noon	74	88	107	116	121	120	116	115	103	87	73	68	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.11	3.02	4.25	5.42	6.49	7.42	7.74	6.71	5.16	3.45	2.32	1.90	(44)	
	RadStd		0.30	0.36	0.50	0.50	0.53	0.33	0.23	0.22	0.21	0.32	0.31	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+0.58	N/A	N/A	N/A	N/A	N/A	+76
(47) Regional (4 neighbors)		N/A	N/A	N/A	+0.71	N/A	N/A	N/A	N/A	N/A	+61

Nomenclature: See separate page